

SAWSAN ABDULBARI

Data Scientist & Machine Learning Engineer | AI Innovation

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BIOGRAPHY

I build full-stack AI and ML systems that ship production machine learning, generative AI, and data software used by real teams to drive measurable business outcomes and automation. Before moving into engineering, I spent over a decade as an operator including 7 years as a founder, plus senior management and quality control roles. That experience shapes how I build AI today: focused on real-world impact, adoption, and audit-ready governance rather than accuracy in isolation. **Trilingual** (Arabic, English, Finnish) with an entrepreneurial background and a focus on climate-aligned solutions.

Professional overview: Contributed to environmental sustainability by transforming plastic waste into valuable resources; managed supply operations and new business initiatives aligned with climate mitigation.

Differentiators: global collaboration (15+ contributor teams), production ML deployment, 7-year founder experience, climate sustainability focus, mentoring and curriculum development.

WORK EXPERIENCE

Junior Expert specializing in AI Aug 2026 - Present

Demos Helsinki | Helsinki Metropolitan Area · Hybrid

Applying AI expertise to Connected Knowledge System (CKS), contributing to development and delivery of AI-powered knowledge solutions.

Artificial Intelligence Machine Learning Connected Knowledge System

Data Scientist and Machine Learning Engineer Incoming — anticipated Dec 2026

University of Helsinki | Helsinki, Finland

Incoming role: data science and machine learning engineering at the University of Helsinki

Data Science Machine Learning Python

Data Scientist and Machine learning Engineer Jun 2026 - Aug 2026

Dashtera | Finland

Built and deployed data science and ML solutions with end-to-end workflows for Dashtera products and services.

Data Science Machine Learning Python

Machine Learning Engineer 2024 - Present

Omdena | Palo Alto, CA (Remote)

Collaborating on AI innovation in globally distributed teams. Researched and implemented RAG architecture for enterprise applications.

Python RAG Architecture Machine Learning AI Research

Student Assistant 2024 - 2025

Häme University of Applied Sciences (HAMK) | Finland

Enhanced outcomes for 50+ students with curriculum support and personalized technical mentoring in Python/data science.

Python Teaching Curriculum Development

Lead Machine Learning Engineer 2024

Omdena Kitwe, Zambia | Remote

COVID-19 forecasting for the Kitwe, Zambia chapter: XGBoost models, Streamlit app, and MAE/RMSE/R² in the public repository. Global team of 15+ contributors.

Team Leadership XGBoost Healthcare AI

Machine Learning Engineer 2023 - 2024

Raqameyyah Digital Consultancy | Egypt (Remote)

Built and deployed ML models; developed chatbots with prompt engineering and front-end applications for customer reach.

Machine Learning Feature Engineering Application Deployment Python

Data Analyst 2023

Raqameyyah Digital Consultancy | Egypt (Remote)

Cleaned and preprocessed datasets; built filtering/visualization apps and automated data processing pipelines (Streamlit).

Data Analysis Dashboards Streamlit EDA

Founder 2007 - 2014

Plastic Recycling for Sustainable Solutions | Hybrid

Founded and managed a plastic recycling factory: business strategy, operations, team leadership, and converting plastic waste into valuable resources.

Business-to-Business (B2B) New Business Development Supply Operations Environmental Sustainability

Earlier roles (Finland / Cyprus) 2021 - 2022

foodora · Foody Allen · BSB Fashion · Keinuipuiston kukka

Accounting internship (foodora); social media and event content; fashion marketing/e-commerce; retail sales and online shop operations including Foodora/Wolt delivery channels.

Excel E-Commerce Marketing Customer Service

EDUCATION

Bachelor of Business Information Technology (Data Science and Machine learning) — Grade 4.7
Häme University of Applied Sciences (HAMK), Finland · 2023–2026

Specialization in Data Science, AI and Business Intelligence. **Thesis:** Microsoft-based performance goal management system for strategy implementation. Catalogue: HAMK Finna (vanaicat.144542).

Climate Change AI (CCAI) Summer School · 2024

Intensive program on machine learning and AI for climate change mitigation, adaptation, and environmental decision-making.

Google Data Analytics Specialization · Nov 2023

Foundations through capstone case study; certificate available on the portfolio site.

Vocational Qualification in Business (Business Administration) · Tredu, Finland · 2020–2022

Business administration fundamentals, project management, and organizational skills.

LANGUAGES

Arabic	English	Finnish
Native	Professional	Professional

PROJECT REFERENCES

Portfolio — full project collection with live demos

Featured: Trading Companion (Vite/React/Supabase) · DataDiwan / DataDiwan OS · Omdena COVID forecast (XGBoost metrics) · Smart RAG chatbot · AI Recipe Finder · PIRHA rehabilitation guide · Emotion recognition (EfficientNet) · Carbon fraud detection network.

SKILLS

Machine Learning • Predictive Modeling • Time Series Analysis • RAG Architecture • Model Deployment • Prompt Engineering	Data Science • Data Mining & Analytics • EDA & Visualization • Feature Engineering • Statistical Analysis • Dashboard Creation
Leadership • Team collaboration (15+ people) • Cross-cultural Collaboration • Mentoring & Training • Project Management • Stakeholder Communication	Business • Strategic Planning • Process Optimization • Financial Management • Business Development • Operations Management

SELECTED PUBLICATIONS

Microsoft-based Performance Goal Management System for Strategy Implementation — Bachelor's thesis, Häme University of Applied Sciences (2026). Catalogue: HAMK Finna (vanaicat.144542). PDF on the portfolio site.

PERSONAL INTERESTS

Hiking · Fitness · Food exploration · Continuous learning · Environmental sustainability